

MODEL QUESTION PAPER
COMPUTER APPLICATIONS IN MANAGEMENT - SET I

(Maximum marks: 75)

(Time: 3 hours)

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 marks)

1	Define Memory unit	M 1.02	R
2	Recall System software	M 1.03	R
3	State the term Booting	M1.02	R
4	Write the meaning of buffering	M2.03	A
5	Define operating system	M2.01	R
6	Recollect data communication	M3.01	R
7	Define data processing	M3.02	R
8	Recall Computer networks	M4.01	R
9	Write the meaning of search engines	M4.03	A

PART B

II Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24marks)

1	Outline the architecture of a computer system.	M 1.02	U
2	Summarise the relation between hardware and software.	M 1.01	U
3	List out the classification of printers with example	M 2.01	U
4	Outline the layers of operating system	M 2.02	U
5	Explain batch processing operating system	M 2.01	U
6	Generalize the forms of modulation	M3.04	A
7	Differentiate analog and digital signals	M 3.02	U
8	Summarise types of data processing	M3.01	A
9	List out the advantages of networking	M4.01	A
10	Identify the merits of ring topology	M 4.02	U

PART C

Answer all questions. Each question carries seven marks

(6x7=42 marks)

III	Describe the System development life cycle.	M 1.04	U
	OR		
IV	Explain flowchart and advantages in detail.	M 1.04	U
V	Determine the functions of operating system	M2.03	U
	OR		
VI	Explain the types of application software	M2.03	U
VII	Identify the Components of DBMS	M3.03	U
	OR		
VIII	Describe the Bounded data transmission media	M3.04	A

IX	Recognize the file types in detail.	M3.01	U
	OR		
X	Describes file utilities.	M3.03	U
XI	Explain star topology, list out the pros and cons of it	M4.02	A
	OR		
XII	Summarize the advantages of www in detail.	M4.03	U
XIII	Explain different types of network	M4.01	U
	OR		
XIV	Describe the types of search engines.	M4.03	U

Question Wise Analysis

Course :Computer Applications in Management

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.02	R	1	1 min
I.2	M 1.03	R	1	1 min
I.3	M1.02	R	1	1 min
I.4	M2.03	A	1	1 min
I.5	M2.01	R	1	1 min
I.6	M3.01	R	1	1 min
I.7	M3.02	R	1	1 min
I.8	M4.01	R	1	1 min
I.9	M4.03	A	1	1 min
II.1	M 1.02	U	3	6 min
II.2	M 1.01	U	3	6 min
II.3	M 2.01	U	3	6 min
II.4	M 2.02	U	3	6 min

II.5	M 2.01	U	3	6 min
II.6	M3.04	A	3	6 min
II.7	M 3.02	U	3	6 min
II.8	M3.01	A	3	6 min
II.9	M4.01	A	3	6 min
II.10	M 4.02	U	3	6 min
III	M 1.04	U	7	20 min
IV	M 1.04	U	7	20 min
V	M2.03	U	7	20 min
VI	M 2.03	U	7	20 min
VII	M 3.03	U	7	20 min
VIII	M3.04	A	7	20 min
IX	M 3.01	U	7	20 min
X	M3.03	U	7	20 min
XI	M4.02	A	7	20 min
XII	M4.03	U	7	20 min
XIII	M4.01	U	7	20 min
XIV	M4.03	U	7	20 min

Course :Computer Applications in Management

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	12	34	2	2	2	6	2	14	6	22
2	10	29	3	3	3	9	2	14	8	26
3	11	31	2	2	3	9	4	28	9	39
4	10	29	2	2	2	6	4	28	8	36
Total	43	123	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	7	6
Understanding	91	74
Applying	53	20
Analysing		
Evaluating		
Creating		
Total	123	100

SCHEME OF VALUATION

(Scoring Indicators)- Set I

Computer Applications in Management

PART A

1. Memory Unit is that component of a computer system, which is used to store the data, Instructions and information before, during and after the processing by ALU.
2. Software that is required to control the working of hardware and aid in effective execution of a general user's applications are called System Software.
3. When we switch on the computer, the instructions stored in ROM are automatically executed. These instructions help the computer to load the operating system from external storage device (disk) to internal storage (RAM). This process of loading of operating System from disk to RAM is called **booting**.
4. **Buffering** is a method of overlapping input, output and processing of a single job.
5. An **Operating System (OS)** is an integrated set of programs that is used to manage the various resources and overall operations of a computer system.

6. Data Communication is the process of transferring messages and data from one point to another.

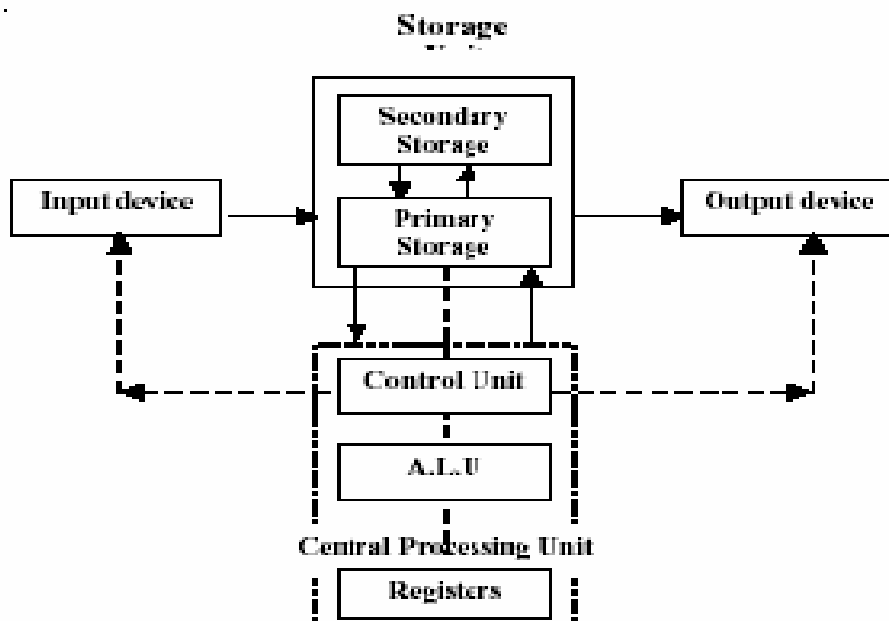
7. **Data processing** is a series of actions or operations that converts data into Information.

8. A **computer network** is a collection of computers and devices connected to each other. The network allows computers to communicate with each other and share resources and information

9. A search engine is a software system that is designed to carry out web searches, which means to search the World Wide Web in a systematic way for particular information specified in a textual web search query.

PART B

1.



2.

1. Both hardware and software are necessary for a computer to do useful job. Both are complementary to each other.

2. Same hardware can be loaded with different software to make a computer perform different types of jobs just as different songs can be played using the same cassette player.

3. Except for upgrades (like increasing main memory and hard disk capacities, or adding speakers, modems, etc.); hardware is normally a one-time expense,

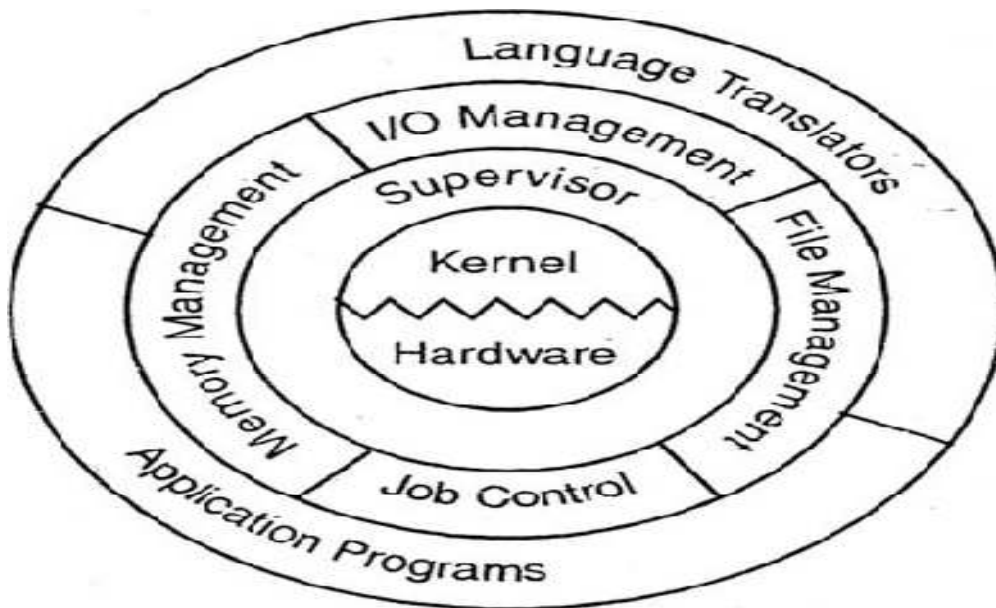
whereas software is a continuing expense. Like we buy new cassettes for newly released songs or for songs whose cassettes, we do not have, we buy, new software to be run on the same hardware as and when need arises, or funds become available.

3. PRINTER-Impact AND Non Impact

IMPACT PRINTERS:-Example: Dot Matrix, Daisy Wheel

NON IMPACT PRINTER-Example: Inkjet, Laser, Thermal, LED

4.



5. Batch processing operating systems are good at churning through large numbers of repetitive jobs on large computers. Jobs like: printing of invitations for AGM, Consolidation of marks and presenting result, working out the pay of each employee in large firm; or processing all the questionnaire forms in a large survey.

In a **batch processing operating system** environment users submit jobs to a central place where these jobs are collected into a batch, and subsequently placed on an input queue at the computer where they will be run. In this case, the user has no interaction with the job during its processing, and the computer's response time is the turnaround time-the time from submission of the job until execution is complete, and the results are ready for return to the person who submitted the job.

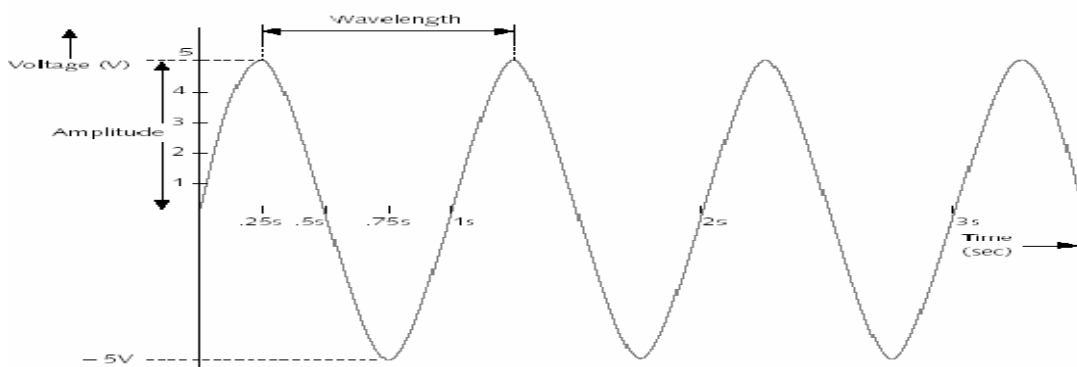
6. There are three forms of modulation- amplitude, frequency and phase modulation

1) **Amplitude modulation (AM)** is a technique used in electronic communication, most commonly for transmitting information via a radio carrier wave. AM works by varying the strength of the transmitted signal in relation to the information being sent.

2) Frequency Modulation (FM) In telecommunications, **frequency modulation (FM)** conveys information over a carrier wave by varying its frequency (contrast this with amplitude modulation, in which the amplitude of the carrier is varied while its frequency remains constant).

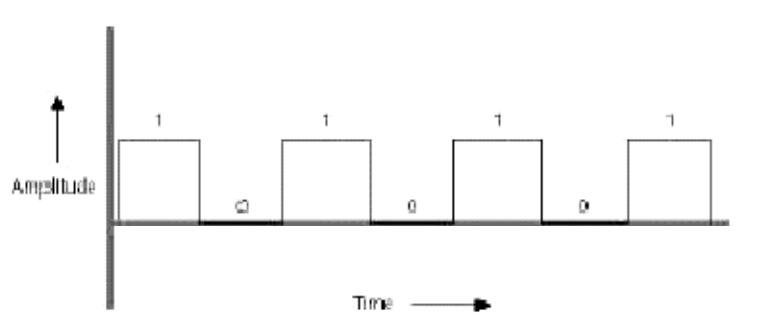
3) Phase modulation (PM) It is a form of modulation that represents information as variations in the instantaneous phase of a carrier wave. Unlike its more popular counterpart, frequency modulation (FM), PM is not very widely used. This is because it tends to require more complex receiving hardware and there can be ambiguity problems in determining whether, for example, the signal has changed phase by $+180^\circ$ or -180° . This scheme can be used with analog or digital data

7. 1. Analog Signaling:- In analog signals, voltage varies continuously. In digital signals, voltage turns off and on repeatedly, pulsing from zero voltage to a specific positive voltage. An analog signal's voltage appears as a continuous wave when graphed over time, because voltage is varied and imprecise in analog signals, analog transmission is more susceptible to transmission flaws such as noise than digital signals.



2. Digital Signaling

Unlike analog signals where there is a smooth curve, digital signals jump directly to the next value.



8. 1. Batch Processing: With batch processing, changes and queries to file are stored for a period of time, and then a processing run is made periodically to update the file and to produce responses to the queries. Batch runs can be made on a scheduled basis, such as daily, weekly, or monthly, or they can be made on an as required basis.

2. Online or immediate processing: In immediate processing, transactions are processed to update the file immediately or shortly after a real-world event occurs. Information processing applications that use immediate processing are often called real time applications. A real time application can immediately capture data about ongoing

events or processes and provide the information necessary to manage them. An airline-reservations system is an example of a real time application.

9.Speed:

Cost:

Centralized Software Management:

Resource Sharing:

Flexible Access:

Security

Requires Administrative Time

File Server May Fail:

10. Benefits of Ring Topology

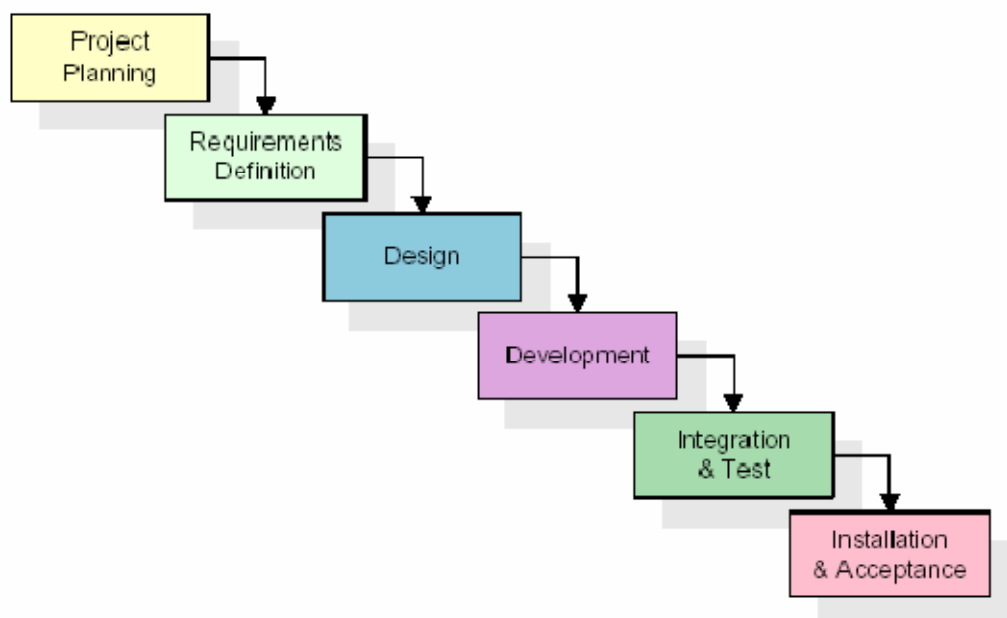
1. Each repeater duplicates the data signals so that very little signal degradation occurs.
2. Ring network works well where there is no central node for making routing decisions.
3. It is more reliable because communication is not dependant on a single central node. If a link between any two nodes fails, or if one of the nodes fails, alternate routing is possible

PART C

III. System development life cycle

Software Development Life Cycle or SDLC is a model of a detailed plan on how to create, develop, implement and eventually fold the software. It's a complete plan outlining how the software will be born, raised and eventually be retired from its function. All software needs to be developed by someone. Developing software and putting it to use is a complex process involving following steps:

1. Analyzing the problem at hand, and planning the program (s) to solve the problem;
2. Coding the program (s);
3. Testing, debugging, and documenting the program (s);
4. Implementing the program (s);
5. Evaluating and maintaining the program (s).



IV. Flowchart and advantages

The following benefits may be obtained from flow charts:

1. Better Communication
2. Effective analysis
3. Effective synthesis
4. Proper program Documentation
5. Effective coding
6. Systematic debugging
7. Systematic testing

V. functions of operating system

An operating system performs, basically the following functions:

1. Communication with the computer user or operator by means of terminals or Consoles, and through the use of monitor commands and responses. It acts as system start up device also. The operating system is automatically loaded into hardware's main memory (RAM) when computer is turned on. Then the OS either waits for you to give it a command or it brings a pre selected program into a ready to operate condition. This sequence is known as boot strapping or simply booting.
2. Assigning Priorities: Usually, the OS has several levels of priorities. For instance, a report generating program may continue to run until the system receives a command or query from the terminal, (a priority in computer lingo). Then the system responds to the request from the terminal before continuing with the report program.
3. The scheduling and loading of programs, or subprograms is necessary in order to provide a continuous job processing sequence or to provide appropriate responses to events- job to job - processing and job as counting.
4. Control over hardware resources e.g. control over the selection and operation of all peripheral devices used to input, or storage. The operating system also directs the movement and processing of information required by your applications programs. It transfers data to and from storage and input output devices like terminals or printers.
5. Managing the data and program files: The operating system directs the information storage and retrieval functions using one or more filing methods. These may include sequential (magnetic tape), random (as on disk, units) or data base storage structures. The latter method uses special identifiers for each file to locate information more rapidly and to avoid- redundant information storage.
6. Provide utility functions that provide the user with tools for some operations like:
 - Loading programs
 - Transferring files from one floppy to another, backing up and copying.
 - Formatting a disk to accept data Program.
 - Sending information to I/O devices like printer, or a modem.
 - Displaying the directory of contents of a disk.
7. Handling errors when they occur and using corrective routines where possible.
8. Protecting hardware, software and data from improper use.

VI. Types of application software

Application software can be further classified into following two types:

- General Purpose Application Software (Database Management Packages, Word Processors, Spreadsheets etc.)

- Special Purpose Application Software (Accounting, Inventory, Production Management etc.)

VII. components of DBMS

A DBMS allows users to organize, process, and retrieve selected data from a database, without any need to know about underlying database structure (organization and location of data). Four major components of a DBMS are:

1. Data Definition Language (DDL)
2. Data Manipulation Language (DML)
3. Query Language
4. Report Generator

Data definition language (DDL) is used to define the structure of a database

Data manipulation language (DML) includes commands to enable users to enter and manipulate data. With these commands, users can add new records to a database, navigate through existing records, view contents of various fields of a record, modify contents of one or more fields of a record, delete an existing record, and sort records in desired sequence.

A query language can be easily learnt by a non-programmer. This enables normal users of a database to obtain desired information from the database without the help of any programmer

Report generators enable users of a database to design the layout of a report in desired format.

VIII. Bounded data transmission media

It is also known as guided media. Bounded transmission media constrain and guide communication signals.

Three common types of bounded media are in use out in the world they are

- a) Coaxial
- b) Twisted pair
- c) Fiber optic

Coaxial cable consists of a central copper core surrounded by an insulator, a braided metal shielding, called braiding, and an outer cover, and called the sheath or jacket. The copper core carries the electromagnetic signal, and the braided metal shielding acts as both a shield against noise and a ground for the signal. The insulator layer usually consists of a plastic material such as polyvinyl chloride (PVC) or Teflon. It protects the copper core from the metal shielding, because if the two made contact, the wire would short-circuit.

b) Twisted-Pair Cable

The most popular network cabling right now is *twisted pair*. It is lightweight, easy to install, inexpensive, and supports many different types of networks. It can also support speeds of up to 100Mbps. Twisted-pair cabling is made up of pairs of solid or stranded copper twisted around each other.

iii. Fiber-Optic Cable

Fiber-optic cable, or simply *fiber*, contains one or several glass fibers at its center, or core. Data are transmitted via pulsing light sent from a laser or light-emitting diode (LED) through the central fibers.

IX. File types

Data files are categorized according to the way an application uses them.

A file management system typically supports following types of files:

1. Transaction File: A transaction file stores input data until it can be processed. For example, in a payroll application for generating monthly pay slips of employees, current month's transaction file contains this month's data of each employee, such as details of hours worked, normal and overtime hours, and if piecework is involved, quantity of goods made.

2. Master File: A master file contains all current data relevant to an application. For example, in payroll application mentioned above, master file contains permanent details of each employee (name, address, employee-code, pay-rate, income-tax-rate, etc.), and also current gross-pay-to-date total and tax-paid-to-date total. When payroll program is processed, it consolidates both master and current month's transaction files to generate this month's pay-slips, and updates master file to make it ready for following month's processing.

3. Output File: Some applications use multiple programs for data processing. In such applications, output produced by one program is fed often as input to another program. Hence, output produced by former program is stored in a file known as output file, which is used later by the latter program.

4. Report File: A report file holds a copy of a report generated by a data processing application in computer-accessible form. A report file can be printed to obtain hardcopy of the report whenever desired. It is advantageous to keep report files instead of paper documents because files are easier to store and carry.

5. Backup File: A backup file is a copy of a file created as a safety precaution against loss of data due to corruption or inadvertent deletion of original file. Regular creation of backup files is extremely important.

X. file utilities

File utilities are routines to perform generalized operations on data files. Normally, they are quite general in nature in the sense that they can deal with different data formats and different types of storage medium. Operations performed by some commonly used file utilities are described below.

Sorting: A file sorting utility is used to arrange records of a file in some defined sequence. Values of certain specified fields (known as *keys*) in each record determine this sequence. The simplest case of sorting is ordering of records in a file on a single key.

Searching: A file searching utility is used to find a particular record in a file. Searching is carried out by matching the values of certain specified fields (known as *keys*) in each record with desired values.

Merging: A file merging utility is used to combine records of two or more ordered (sorted) files into a single ordered file. Records of each of the input files must be sorted in same order, although their record layout need not be identical. A merging utility places records from each of the input files in their correct relative order, producing an output file having all records in same order as input files.

Copying: A file copying utility is used to produce a copy of a file either from one unit of a storage device to another similar unit (such as from one tape reel or floppy disk to another), or from one storage medium to another (such as from tape to hard disk, or from CD-ROM to hard disk). File copying utilities are used often to take back-up copies of files.

Printing: A file printing utility is used to print a file on a printer to produce hard copy of its contents. Printing utilities often provide facility to print file contents in different formats.

Maintenance: A file maintenance utility is used to copy data from one or more files to a new file selectively, or to update a file's contents selectively. For example, a file maintenance utility may provide features to combine data from more than one file into a single file, delete records in a file identified by record key values or record count, and select specific fields of records to be copied from an existing file to a new file.

XI. Explain star topology, pros and cons

Star Topology: Many home networks use the star topology. A star network features a central connection point called a "hub" that may be a hub, switch or router. Devices typically connect to the hub with Unshielded Twisted Pair (UTP) Ethernet. The star topology is a popular method of connecting the cabling in a computer network. In a star, each device connects to a central point via a point-to-point link. Depending on the logical architecture used, several names are used for the central point including the following:

- Hub
- Multipoint Repeater
- Concentrator
- Multi-Access Unit (MAU)

Compared to the bus topology, a star network generally requires more cable, but a failure in any star network cable will only take down one computer's network access and not the entire LAN. (If the hub fails, however, the entire network also fails).

Benefits of Star Topology

1. Each device is isolated on its own cable. This makes it easy to isolate individual devices from the network by disconnecting them from the wiring hub.
2. All data goes through the central point, which can be equipped with diagnostic devices that make it easy to trouble shoot and manage the network.
3. Hierarchical organization allows isolation of traffic on the channel. This is beneficial when several, but not all, computers place a heavy load on the network. Traffic from those heavily used computers can be separated from the rest or dispersed throughout for a more even flow of traffic.
4. It has minimal line cost because only $n-1$ lines are required for connecting n nodes.

Disadvantages of Star topology

1. Because point-to-point wiring is utilized for each node, more cable is required.
2. Hub failures can disable large segments of the network.

XII. Advantages of www

Standard Interface: Because the web presents a standard interface to FTP, News, e-mail, gopher and other Internet tools, as well as a standard way of accessing information through web pages and databases through forms, the web has become an invaluable tool for simple access to the net.

• **Formatted Output:** The Web supports formatted text. You can have:

□ **Headings of a number of sizes...**

while normal text can be formatted in a number of ways, including: **bold**, *italicized*, super or sub-scripted. You can also have lists and tables.

□ **Hypertext:** As mentioned above, web pages use hypertext links to provide a very easy and straightforward way to navigate around the net. Hypertext links can be within a document, between web pages on the same computer or between web pages on different computers. Hypertext links can also be between graphics, audio and video.

□ **Multimedia:** One of the big advantages the web over earlier systems such as telnet-based programs and gopher is that it is multimedia capable.

Any of the graphical browsers (like Internet Explorer and Netscape Navigator) can display formatted text, as well as GIF images. Some browsers can also handle JPEG images.

□ **Interactive Pages:** With **forms**, users can interact with web pages to obtain dynamic or customized information. A typical example of form is the registration form that you have to fill up while registering for an email account.

XIII. Types of network

LAN (Local Area Network)

WAN (Wide Area Network)

MAN (Metropolitan Area Network)

Local Area Network (LAN)

LAN stands for **Local Area Network**. It's a group of computers which all belong to the same organization, and which are linked within a small geographic area using a network, and often the same technology (the most widespread being Ethernet). This is a computer network covering a small physical area, like a home, office, or small group of buildings, such as a school, or an airport

A local area network is a network in its simplest form. Data transfer speeds over a local area network can reach up to 10 Mbps (such as for an Ethernet network) and 1 Gbps (as with FDDI or Gigabit Ethernet). A local area network can reach as many as 100, or even 1000, users. By expanding the definition of a LAN to the services that it provides, two different operating modes can be defined:

- In a "**peer-to-peer**" network, in which communication is carried out from one computer to another, without a central computer, and where each computer has the same role.
- In a "**client/server**" environment, in which a central computer provides network services to users.

Metropolitan Area Network (MAN)

A Metropolitan Area Network is a system of LANs connected throughout a city or metropolitan area. A **metropolitan area network** (MAN) is a network that connects two or more local area networks or campus area networks together but does not extend beyond the boundaries of the immediate town/city. Routers, switches and hubs are connected to create a metropolitan area network. MANs have the requirement of using telecommunication media such as voice channels or data channels. Branch offices are connected to head offices through MANs. Examples of organizations that use MANs are universities and colleges, grocery chains, and banks. The main criterion for a MAN is that the connection between LANs is through a local exchange carrier (the local phone company).

WAN (Wide Area Network)

As the term implies, a **WAN** spans a large physical distance. The Internet is the largest WAN, spanning the Earth. A WAN is a geographically-dispersed collection of LANs. A network device called a router connects LANs to a WAN. In IP networking, the router maintains both a LAN address and a WAN address. A WAN differs from a LAN in several important ways. Most WANs (like the Internet) are not owned by any one organization but rather exist under collective or distributed ownership and management.

A Wide Area Network is a network system connecting cities, countries, or continents together. WANs are connected together using one of the telecommunications media. The main difference between a MAN and a WAN is that the WAN uses Long Distance Carriers. Otherwise the same protocols and equipment are used as a MAN.

XIV. Types of search engines

Search engines are automated computers owned by companies such as Google, AltaVista, Yahoo etc that spend all of their time day and night surfing the Internet.

A search engine helps you find information on the Internet. To use a search engine, you must enter a word (or words) that you expect will be in the Web pages or newsgroup messages you're looking for, then click a Search button.

Crawler-Based Search Engines: Crawler-based search engines, such as Google, create their listings automatically. They "crawl" or "spider" the web, then people

search through what they have found. If you change your web pages, crawler-based search engines eventually find these changes, and that can affect how you are listed. Page titles, body copy and other elements all play a role.

- **Human-Powered Directories:** A human-powered directory, such as the Open Directory, depends on humans for its listings. You submit a short description to the directory for your entire site, or editors write one for sites they review. A search looks for matches only in the descriptions submitted. Changing your web pages has no effect on your listing. Things that are useful for improving a listing with a search engine have nothing to do with improving a listing in a directory. The only exception is that a good site, with good content, might be more likely to get reviewed for free than a poor site.

- **Hybrid Search Engines:** In the web's early days, it used to be that a search engine either presented crawler-based results or human-powered listings. Today, it is extremely common for both types of results to be presented. Usually, a hybrid search engine will favor one type of listings over another.

MODEL QUESTION PAPER

COMPUTER APPLICATIONS IN MANAGEMENT - SET II

(Maximum marks: 75)

(Time: 3 hours)

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 marks)

1	Define Primary Memory	M 1.02	U
2	State the term Software	M 1.03	U
3	Recall Algorithm	M1.04	R
4	Write the meaning of spooling	M2.01	U
5	Define Multitasking	M2.02	U
6	Recall Data base	M3.03	U
7	Define Data modulation	M3.02	R
8	Write the meaning of Network topology	M4.01	R
9	State the term 'www'	M4.03	R

PART B

II Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 marks)

1	List out the special input devices	M 1.02	U
2	Explain types of RAM	M 1.02	U

3	Outline basic flow charting shapes	M 1.04	U
4	Identify the characteristics of an operating system	M 2.01	U
5	Explain time sharing operating system	M 2.02	A
6	Describe QBE	M3.03	U
7	Explain DBMS with Examples	M 3.01	U
8	Enumerate the bounded media	M3.04	A
9	Summarize Advantages of bus topology	M4.02	A
10	List out the Merits of e mail.	M 4.02	U

PART C

Answer all questions. Each question carries seven marks

(6x7=42 marks)

III	Describe the Characteristics of computer .	M1.01	R
	OR		
IV	Explain Types of memory in detail.	M1.02	U
V	Summarize operating system techniques.	M2.01	U
	OR		
VI	Explain Types of operating system.	M2.01	U
VII	Describe database models.	M3.01	U
	OR		
VIII	Generalize unbounded data transmission media.	M3.04	A
IX	Explain data transmission modes.	M3.04	A
	OR		
X	Summarize the pros and cons of digital signal.	M3.03	U
XI	Describe computer network can help your business.	M 4.01	U
	OR		
XII	Explain bus topology. List out its pros and cons.	M4.02	A
XIII	Describe network topology	M4.02	A
	OR		
XIV	Summarize the advantages of networking	M4.03	U

Question Wise Analysis

Q. No	Module Outcome	Cognitive Level	Marks	Time
I.1	M 1.02	U	1	1 min
I.2	M 1.03	U	1	1 min
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I.6	M3.03	U	1	1 min
I.7	M3.02	R	1	1 min
I.8	M4.01	R	1	1 min
I.9	M4.03	R	1	1 min
II.1	M 1.02	U	3	6 min
II.2	M 1.02	U	3	6 min
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II.5	M 2.02	A	3	6 min
II.6	M3.03	U	3	6 min
II.7	M 3.01	U	3	6 min
II.8	M3.04	A	3	6 min
II.9	M4.02	A	3	6 min
II.10	M 4.02	U	3	6 min
III	M 1.01	R	7	20 min
IV	M 1.02	U	7	20 min
V	M2.01	U	7	20 min
VI	M 2.01	U	7	20 min
VII	M 3.01	U	7	20 min
VIII	M3.04	A	7	20 min
IX	M 3.04	A	7	20 min
X	M3.03	U	7	20 min

XI	M4.01	U	7	20 min
XII	M4.02	A	7	20 min
XIII	M4.02	A	7	20 min
XIV	M4.03	U	7	20 min

Course :Computer Applications in Management

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	12	34	3	3	3	9	2	14	8	26
2	10	29	2	2	2	6	2	14	6	22
3	11	31	2	2	3	9	4	28	9	39
4	10	29	2	2	2	6	4	28	8	36
Total	43	123	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	11	9
Understanding	66	54
Applying	46	37
Analysing		
Evaluating		
Creating		
Total	123	100

SCHEME OF VALUATION (Scoring Indicators)- Set II

Computer Applications in Management

Part A

1. Memory- is also known as the primary storage or main memory/ internal memory. It is a part of microcomputer that holds data for processing, instructions for processing the data (the program) and information (processed data)
2. **software** refers to a set of computer programs, procedures, and associated documents (flowcharts, manuals, etc.) describing the program, and how they are to be used.
3. **Algorithm** refers to the logic of a program. It is a step – by – step description of how to arrive at a solution.
4. spooling is a specialized form of multiprogramming for the purpose of copying data between different devices.

5. It refers to the ability of an operating system to execute two or more tasks concurrently. In multitasking environment, the user opens new applications without closing the previous ones and the information can be easily moved among a number of applications.
6. A database is a collection of structured data. The structure of the data is independent of any particular application.
7. Data modulation is a technology used to modify analog signals in order to make them suitable for carrying data over a communication path.
8. Topology of a network refers to the way in which the network's nodes (computers or other devices that need to communicate) are linked together. It determines the various data paths available between any pair of nodes in the network.
9. World Wide Web is a system of Internet servers that uses HTTP to transfer specially formatted documents.

PART B

1. Special Input Devices

The input devices, which are not essential to operate a PC, are called as Special Input Devices. These devices are used for various special purposes and are generally not required for basic input operations. These devices include Trackball, Light Pen, TouchScreen, Joystick, Digitizer, Scanner, OMR, OCR, Bar Code Reader, MICR and Voice Input Devices.

2. Types of RAM

There are two types of RAM used in PCs - Dynamic and Static RAM.

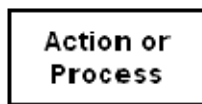
Dynamic RAM (DRAM): The information stored in Dynamic RAM has to be refreshed after every few milliseconds, otherwise it is erased. DRAM has higher storage capacity and is cheaper than Static RAM.

Static RAM (SRAM): The information stored in Static RAM need not be refreshed, but it remains stable as long as power supply is provided. SRAM is costlier but has higher speed than DRAM.

3. Flowcharts use special shapes to represent different types of actions or steps in a process. Lines and arrows show the sequence of the steps, and the relationships among them.



The terminator symbol marks the starting or ending point of the system. It usually contains the word "Start" or "End."



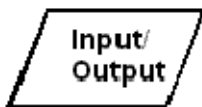
A box can represent a single step ("add two cups of flour"), or and entire subprocess ("make bread") within a larger process.



A printed document or report.



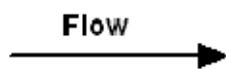
A decision or branching point. Lines representing different decisions emerge from different points of the diamond. The box where the computer can make decisions like finding out which numbers is bigger or smaller or equal than another number, depending on the information it receives. These are diamond-shaped boxes. The computer can decide with the help of decision taken by the diamond box, which route is to be followed further.



Represents material or information entering or leaving the system, such as a customer order (input) or a product (output).



Indicates that the flow continues on another page, where a matching symbol (containing the same letter) has been placed.



Lines indicate the sequence of steps and the direction of flow.

4. The main characteristics of an operating system:

- Concurrency: being able to handle events as they occur and executing multiple tasks in parallel;
- Sharing the resources for a number of reasons:

1. Cost

2. Using the work of others

3. Sharing data: use the same data in several different programs possibly used by several different users.

- Removing redundancy;

- Long-term storage of information (privacy, integrity, security);
- Non determinacy: an operating system must be able to handle events occurring in an unpredictable order.

5. Time-sharing is a mechanism to provide simultaneous interactive use of a computer system by many users in such a way that each one feels that he/she is the sole user of the system. It uses multiprogramming with a special CPU scheduling algorithm to achieve this. A time-sharing system has many (even hundreds of) user terminals connected to the same computer simultaneously. Using these terminals, multiple users can work on the system.

Advantages of Time-sharing Systems: Although time-sharing systems are complex to design, they provide the following advantages to their users:

1. Reduces CPU idle time. A user's thinking and typing speed is much slower than a computer's processing speed. Hence, during interactive usage of a system, while a user is engaged in thinking or typing his / her input, a time-sharing system services many other users. Hence, time-sharing systems help in reducing CPU idle time and, in turn, provide increased system throughput.

2. Provides advantages of quick response time. The special CPU scheduling algorithm used in time sharing systems ensures quick response time to all users. This feature helps in improving programmers' efficiency by making interactive programming and debugging much simpler and quicker. Multiple programmers can work simultaneously for writing, testing, and debugging their programs, or for trying out various approaches to a problem solving.

3. Offers good computing facility to small users. Small users can gain direct access to more sophisticated hardware and software than they could otherwise justify or afford. In time-sharing systems, they merely pay a fee for resources used and are relieved of hardware, software, and personnel problems associated with acquiring and maintaining their own installation.

6. query languages are easy to learn and use, many database developers further simplify the job of specifying search criteria. For this, they provide a simpler user interface (called front end) to query language. A database user specifies facts about a query by using this interface and a query language builder composes query language statements automatically from the facts. Front end usually consists of a form (called *QBE form*), and a user specifies a search criteria simply by inputting values into the fields of this form. Once the user completes the QBE form, QBE engine converts user inputs automatically into suitable query language statement(s) for search processing. Hence, the user is relieved of remembering query language keywords and using them with correct syntax to form queries.

This front-end feature *is* called **queryby example (QBE)**. It is very useful for many database users, especially beginners.

7. A database management system (DBMS) is a program that allows users to define,manipulate, and process the data in a database in order to produce meaningful information.DBMS is collection of programs that enables you to store, modify, and extract importantinformation from a database. There are many different types of DBMS, ranging from smallsystems that run on personal computers to huge systems that run on mainframes.

examples of DBMS:

- Computerized library systems
- Automated teller machines
- Flight reservation systems
- Computerized parts inventory systems

8. It is also known as guided media. Bounded transmission media constrain and guidecommunication signals.

Three common types of bounded media are in use out in the world they are

- a) Coaxial
- b) Twisted pair
- c) Fiber optic

9. **Bus Topology:** Bus networks (not to be confused with the system bus of a computer) use acommon backbone to connect all devices. A single cable, the backbone functions as ashared communication medium that devices attach or tap into with an interface connector.A device wanting to communicate with another device on the network sends a broadcastmessage onto the wire that all other devices see, but only the intended recipient actuallyaccepts and processes the message. Ethernet bus topologies are relatively easy to install and don't require much cabling compared to the alternatives

Benefits of Bus Topology

1. Cabling costs are minimized because of the **common trunk**.
2. Failure of a node does not affect communication among other nodes in the network
3. Addition of new nodes to the network is easy.

10. Advantages of E-Mail

□ **Speed:** Because e-mail is based around computer networks, it is fast - much faster than standard mail (otherwise known as snail mail). An e-mail message that you send to someone in Finland or Brazil could take only a few seconds to reach them.

□ **Cost:** Depending on how you have obtained your Internet access, e-mail is free

or very cheap, much cheaper than the 45 cents it takes to send a standard letter within Australia.

□ **Flexibility:** With the new MIME e-mail system being used today, you aren't limited to text in your e-mail messages. You can include pictures, sound files, movies, data files from your favoritespreadsheet program, etc.

PART C

III. Characteristics of computer

1. Speed: Computer is very fast and accurate device. Since electronic pulses travel

at incredible speed and they are electronic device its internal speed is virtually instantaneous. A microcomputer can process millions of instruction per seconds over and over again without any mistake.

2. Accuracy: Computers physical circuits rarely make errors, if the data and instruction are correctly fed. Most errors which occur in computers are either hardware error or human error

3. Storage: Computers have a large amount of memory to hold a very large amountof data, we can store large amount of data information in the secondary storage device.

4. Programmability: A computer is programmable; i. e. what computer does depend on the lines of instruction (Program) it is using.

5. Diligence: Computer is free from problems like lack of concentration, and confusions etc. Computer is never confused like humans and it can perform instruction again and again without failing or getting bored.

6. Versatility: We can perform many different types of tasks on computer, one moment it might be busy in calculating the statistical date for annual performance

evaluation of a business organization and next moment it might be working on inventory control.

7. Power of remembrance: Unlike humans, computer can store things for unlimitedperiod of time. It has great remembering power.

IV. Types of memory

Memory

Memory- is also known as the primary storage or main memory/ internal memory. It is aapart of microcomputer that hold data for processing, instruction for processing the data(the program) and information (processed data) .It is of the following three types:

1. Read Only Memory (ROM)
2. Random Access Memory (RAM)
3. Complementary Metal Oxide Semiconductor Memory (CMOS)

1. Read Only Memory

Read Only Memory is an essential component of the memory unit. We know that the computer, being a machine, itself has no intelligence or memory and requires

instructions, which are given by a person. Whenever the computer is switched on, it searches for the required instructions. The memory, which has these essential

instructions, is known as Read Only Memory (ROM). This memory is permanent and is

not erased when the system is switched off. As appears with its name, it is read only memory i.e. it can be read only and not be written by user/ programmer. The memory capacity of ROM varies from 64 KB to 256 KB (1 Kilobyte = 1024 bytes) depending on the model of computer.

2. Random Access Memory

Random Access Memory (RAM) is another important component of the Memory Unit. It is used to store data and instructions during the execution of programs. Contrary to ROM, RAM is temporary and is erased when the computer is switched off. RAM is a read/ write type of memory, and thus can be read and written by the user/ programmer. As it is possible to randomly use any location of this memory, therefore, this memory is known as random access memory. The memory capacity of RAM varies from 640 KB to several megabytes (1 Megabyte = 1024 KB) with different models of PC.

3. Complementary Metal Oxide Semiconductor Memory

Complementary Metal Oxide Semiconductor (CMOS) memory is used to store the

system configuration, date, time and other important data. When the computer is switched on, BIOS matches the information of CMOS with the peripheral devices and displays error in case of mismatching.

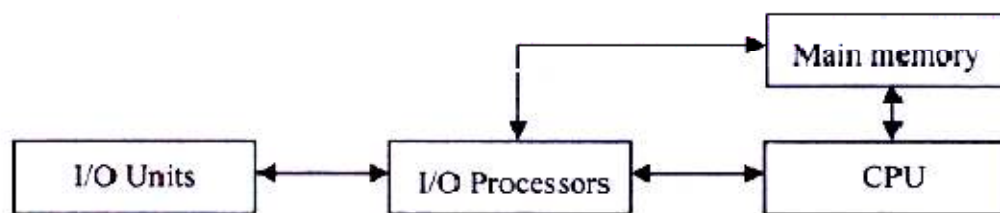
V. operating system techniques

Multiprogramming:- It is a process by which single CPU works on two or more programs simultaneously. Using this technique, the operating system keeps the CPU busy. Multiprogramming allows the processor to handle either multiple batch jobs at a time (Batch Multiprogramming) or multiple interactive jobs

shared among multiple users (Time Sharing Multiprogramming). Time-sharing is a technique that allows a CPU to simultaneously support the activities of several users by allocating fixed time slots (in milliseconds). Examples of operating systems that support multiprogramming are OS/2, UNIX and Macintosh System

Multiprocessing

It refers to the use of two or more CPUs to perform a coordinated task simultaneously.



Multitasking

It refers to the ability of an operating system to execute two or more tasks concurrently. In a multitasking environment, the user opens new applications without closing the previous ones and the information can be easily moved among a number of applications.

VI. Types of operating system.

As computers have progressed and developed so have the types of operating systems. Many types of operating systems are available for computers, which can be divided into the following two broad categories -

- (a) Single-user operating systems.
- (b) Multi-user operating systems.

Modern computer operating systems may also be classified into three other groups, which are distinguished by the nature of interaction that takes place between the computer user and his or her program during its processing. The three groups are called **batch**, **timeshared** and **real time** operating systems.

Batch Processing Operating Systems:-

In a **batch processing operating system** environment users submit jobs to a central place where these jobs are collected into a batch, and subsequently placed on an input queue at the computer where they will be run. In this case, the user has no interaction with the job during its processing, and the computer's response time is the turnaround time - the time from submission of the job until execution is complete, and the results are ready for return to the person who submitted the job.

Time Sharing Operating Systems

Time-sharing is a mechanism to provide simultaneous interactive use of a computer system by many users in such a way that each one feels that he/she is the sole user of the system. It uses multiprogramming with a special CPU scheduling algorithm to achieve this. A time-sharing system has many (even hundreds of) user terminals connected to the same computer simultaneously. Using these terminals, multiple users can work on the system simultaneously.

Real Time Operating Systems

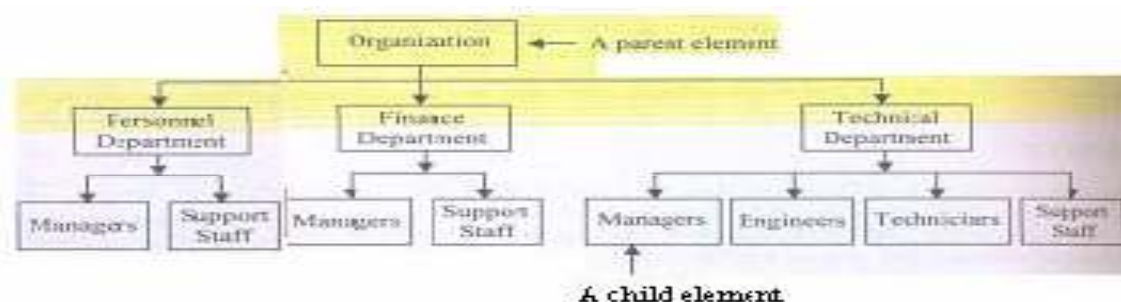
A Real-Time Operating System (**RTOS**) is a multitasking operating system intended for real-time applications. Such applications include embedded systems (programmable thermostats, household appliance controllers), industrial robots, spacecraft, industrial control (see SCADA), and scientific research equipment. It is another form of operating system which is used in environments where a large number of events mostly external to computer systems, must be accepted and processed in a short time or within certain deadlines.

VII. database models

A **database model** defines the manner in which various files of a database are linked together. Four commonly used database models are hierarchical, network, relational, and object-oriented.

Hierarchical Databases

The **hierarchical data model** organizes data in a tree structure. There is a hierarchy of parent and child data segments.



Network Model

In the **network model** of a database it is possible for a record to have multiple parents, making the system more flexible compared to the strict single-parent model of the hierarchical database.

A **Relational database** allows the definition of data structures, storage and retrieval operations and integrity constraints. In such a database the data and relations between them are organized in tables. A table is a collection of records and each record in a table contains the same fields.

Object-Oriented Model:- data exists in the form of objects, which include both the data and the data's behavior. An **object-oriented database** is a collection of objects whose behavior, state, and relationships are defined in accordance with object-oriented concepts (such as object, class, class hierarchy, etc.). An object-oriented database management system allows definition and manipulation of an object-oriented database.

VIII. Unbounded data transmission media

Unbounded, or wireless, media does not use any physical connectors between the two devices communicating. Usually the transmission is sent through the atmosphere, but sometimes it can be just across a room. Wireless media is used when a physical obstruction or distance blocks the use of normal cable media. Following are the types of unbounded media:

1. Radio Waves

- i. Short-wave
- ii. Very-high frequency (VHF) television and radio
- iii. Ultra-high frequency (UHF) television and radio Micro waves

2. Microwaves

- i. Terrestrial Microwaves
- ii. Satellite Microwaves

3. Infrared

- i. Point-to-point
- ii. Broadcast

IX. data transmission modes

Data transmission, whether analog or digital, may also be characterized by the direction in which the signals travel over the media.

1. Simplex:

In cases where signals may travel in only one direction, the transmission is considered simplex.

For example,

A football coach calling out orders to his team through a megaphone is using simplex communication. In this example, the coach's voice is the signal, and it travels in only one direction—away from the megaphone's mouthpiece and toward the team. Simplex is sometimes called one-way, or unidirectional, communication.

2. Half Duplex

In half-duplex transmission signals may travel in both directions over a medium but in only one direction at a time. Half-duplex systems contain only one channel for communication, and that channel must be shared for multiple nodes to exchange information.

3. Full-Duplex

When signals are free to travel in both directions over a medium simultaneously, the transmission is considered full duplex. Full duplex may also be called bi-directional transmission or sometimes, simply duplex.

X. pros and cons of digital signal

Advantages of Digital

1. Less expensive
2. More reliable
3. Easy to manipulate
4. Flexible
5. Compatibility with other digital systems
6. Only digitized information can be transported through a noisy channel without degradation
7. Integrated networks

Disadvantages of Digital

1. Sampling Error
2. Digital communications require greater bandwidth than analogue to transmit the same information.
3. The detection of digital signals requires the communications system to be synchronized, whereas generally speaking this is not the case with analogue systems.

XI. Computer network can help your business

File sharing: A network makes it easy for everyone to access the same file and prevents people from accidentally creating different versions.

Printer sharing: If you use a computer, chances are you also use a printer. With a

network, several computers can share the same printer. Although you might need a more expensive printer to handle the added workload, it's still cheaper to use a network printer than to connect a separate printer to every computer in your office.

Communication and collaboration: It's hard for people to work together if no one knows what anyone else is doing. A network allows employees to share files, view other people's work, and exchange ideas more efficiently. In a larger

office, you can use e-mail and instant messaging tools to communicate quickly and to store messages for future reference.

Organization: A variety of scheduling software is available that makes it possible

to arrange meetings without constantly checking everyone's schedules. This software usually includes other helpful features, such as shared address books and to-do lists.

Remote access: Having your own network allows greater mobility while maintaining the same level of productivity. With remote access in place, users are able to access the same files, data, and messages even when they're not in the office. This access can even be given to mobile handheld devices.

Data protection: You should know by now that it's vital to back up your computer data regularly. A network makes it easier to back up all of your company's data on an offsite server, a set of tapes, CDs, or other backup systems. Of course, another aspect of data protection is data security.

XII. explain bus topology ,prons and cons

Bus Topology: Bus networks (not to be confused with the system bus of a computer) use a common backbone to connect all devices. A single cable, the backbone, functions as a shared communication medium that devices attach or tap into with an interface connector. A device wanting to communicate with another device on the network sends a broadcast message onto the wire that all other devices see, but only the intended recipient actually accepts and processes the message. Ethernet bus topologies are relatively easy to install and don't require much cabling compared to the alternatives.

Benefits of Bus Topology

1. Cabling costs are minimized because of the **common trunk**.
2. Failure of a node does not affect communication among other nodes in the network.
3. Addition of new nodes to the network is easy.

Disadvantages of Bus topology

1. Difficult to trouble shoot because no central distribution points exist.
2. Cable breaks can disable the entire segment because they remove the required termination from each of the two cable fragments.

XIII. Network topology

Network topologies are categorized into the following basic types:

1. Bus Topology;
2. Ring Topology;
3. Star Topology;
4. Tree Topology;
5. Mesh Topology.

Bus Topology: Bus networks (not to be confused with the system bus of a computer) use a common backbone to connect all devices. A single cable, the backbone, functions as a shared communication medium that devices attach or tap into with an interface connector. A device wanting to communicate with another device on the network sends a broadcast message onto the wire that all other devices see, but only the intended recipient actually accepts and processes the message. Ethernet bus topologies are relatively easy to install and don't require much cabling compared to the alternatives.

Ring Topology: In a ring network, every device has exactly two neighbors for communication purposes. All messages travel through a ring in the same direction (either "clockwise" or "counterclockwise"). A failure in any cable or device breaks the loop and can take down the entire network. Ring topologies are found in some office buildings or school campuses.

Star Topology: Many home networks use the star topology. A star network features a central connection point called a "hub" that may be a hub, switch, or router. Devices typically connect to the hub with Unshielded Twisted Pair (UTP) Ethernet. The star topology is a popular method of connecting the cabling in a computer network. In a star, each device connects to a central point via a point-to-point link. Depending on the logical architecture used, several names are used for the central point including the following:

- Hub
- Multipoint Repeater
- Concentrator
- Multi-Access Unit (MAU)

Tree Topology: Tree topologies integrate multiple star topologies together onto a bus. In its simplest form, only hub devices connect directly to the tree bus and each hub functions as the "root" of a tree of devices. This bus/star hybrid approach supports future expandability of the network much better than a bus (limited in the number of devices due to the broadcast traffic it generates) or a star (limited by the number of hub connection points) alone.

Mesh Topology: Mesh topologies involve the concept of routes. Unlike each of the previous topologies, messages sent on a mesh network can take any of several possible paths from source to destination. (Recall that even in a ring, although two cable paths exist, messages can only travel in one direction.) Some WANs, most notably the Internet, employ mesh routing.

XIV. Advantages of networking

Speed: Networks provide a very rapid method for sharing and transferring files. Without a network, files are shared by copying them to floppy disks, then carrying or sending the disks from one computer to another. This method of transferring files in this manner is very time consuming.

Cost: The network version of most software programs are available at considerable savings when compared to buying individually licensed copies. Besides monetary savings, sharing a program on a network allows for easier upgrading of the program. The changes have to be done only once, on the file server, instead of on all the individual workstations.

Centralized Software Management: One of the greatest benefits of installing a network is the fact that all of the software can be loaded on one computer (the file server). This eliminates that need to spend time and energy installing updates and tracking files on independent computers throughout the building.

Resource Sharing: Sharing resources is another area in which a network exceeds stand-alone computers. Most companies cannot afford enough laser printers, fax machines, modems, scanners, and CD-ROM players for each computer. However, if these or similar peripherals are added to a network, they can be shared by many users.

Flexible Access: Networks allow employees/students to access their files from computers throughout the company/college. Students can begin an assignment in their classroom, save part of it on a public access area of the network, and then go to the media center after college to finish their work.

Security: Files and programs on a network can be designated as "copy inhibit," so that you do not have to worry about illegal copying of programs. Also, passwords can be established for specific directories to restrict access to authorized users.

Requires Administrative Time: Proper maintenance of a network requires considerable time and expertise. Many companies have installed a network, only to find that they did not budget for the necessary administrative support.

File Server May Fail: Although a file server is no more susceptible to failure than any other computer, when the file server "goes down," the entire network may come to a halt. When this happens, the entire school may lose access to necessary programs and files.